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Ex-5

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# ~~Experim~~ EXPERIMENT ON PACKET CAPTURE TOOL - WIRESHARK

## AIM:

Experiments on packet capture tool wireshark

## PACKET SNIFFER:

→ Sniffs messages being sent/received from/by your computers

→ Store and display the contents of the various protocol fields in the messages.

→ Passive program

i) Never sends packets itself

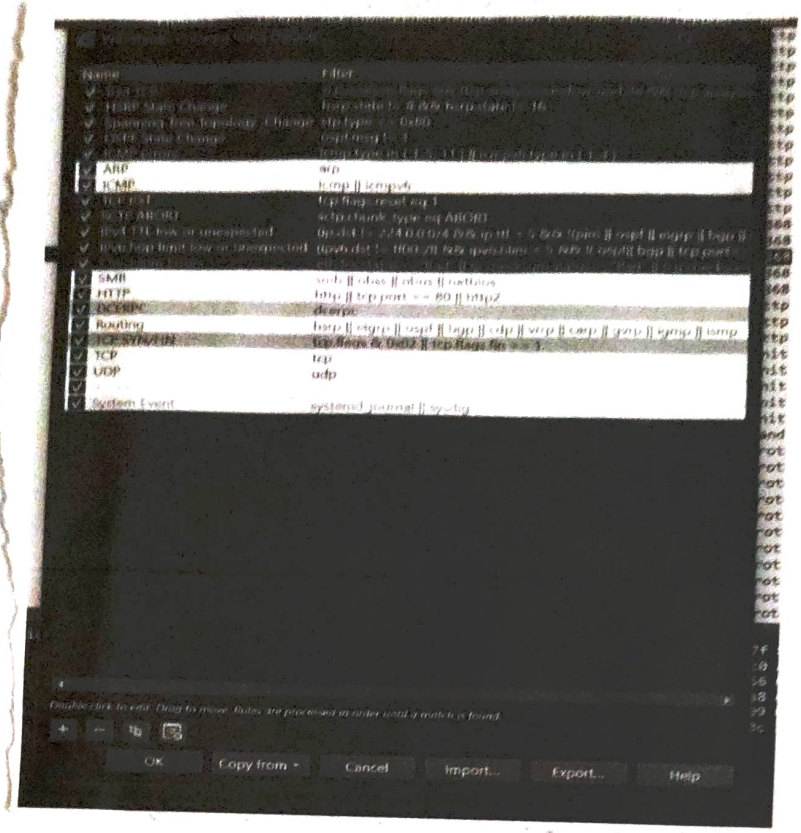
ii) No packets addr to it

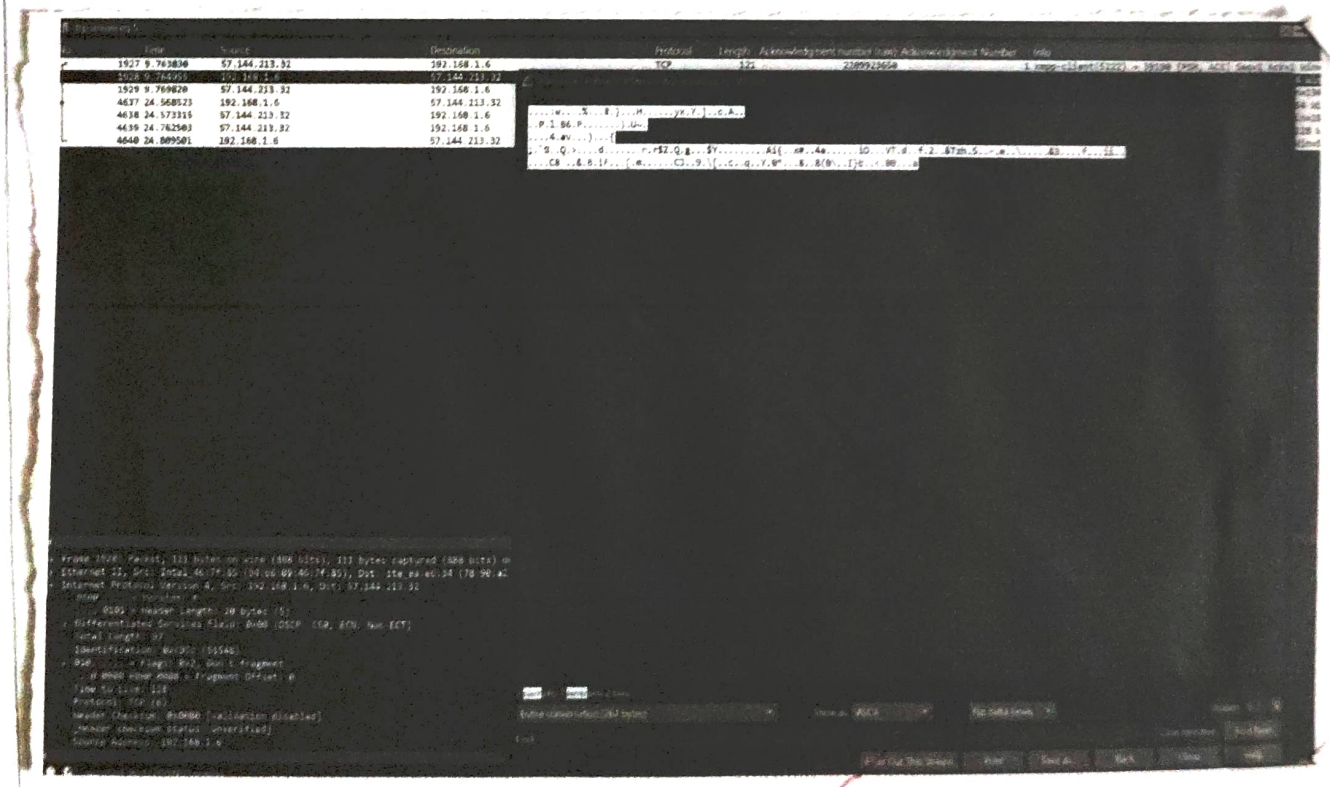
iii) Receivers a copy of all packets (sent/received)

## Packet Sniffer Structure Diagnostic tool:-

* Tcpdump - Eg:- tcpdump -enx host 10.129.41.2 -w exe3.out

* ~~Wireshark~~ - Wireshark -r exe3.out


Name
Filter
ARP
arp
ICMP
icmp || icmpv6
SMB
smb || nbss || nbns || netbios
HTTP
http || tcp.port == 80 || http2
Routing
hsrp || eigrp || ospf || bgp || cdp || vrrp || carp || gvrp || igmp || ismp
TCP
tcp
UDP
udp
System Event
systemd_journal || syslog
OK
Copy from
Cancel
Import...
Export...
Help


No.
Time
Source
Destination
Protocol
Length
Info
1927 9.763830 57.144.213.32 192.168.1.6
1928 9.764950 192.168.1.6 57.144.213.32
1929 9.769820 57.144.213.32 192.168.1.6
4637 24.568523 192.168.1.6 57.144.213.32
4638 24.573315 57.144.213.32 192.168.1.6
4639 24.762503 57.144.213.32 192.168.1.6
4640 24.809501 192.168.1.6 57.144.213.32

PROCEDURE:-

→ Select the local area connection in Wireshark

→ Go to capture option

→ Select capture and stop at automatically after 100 packets

→ Then click start capture

→ Save packets

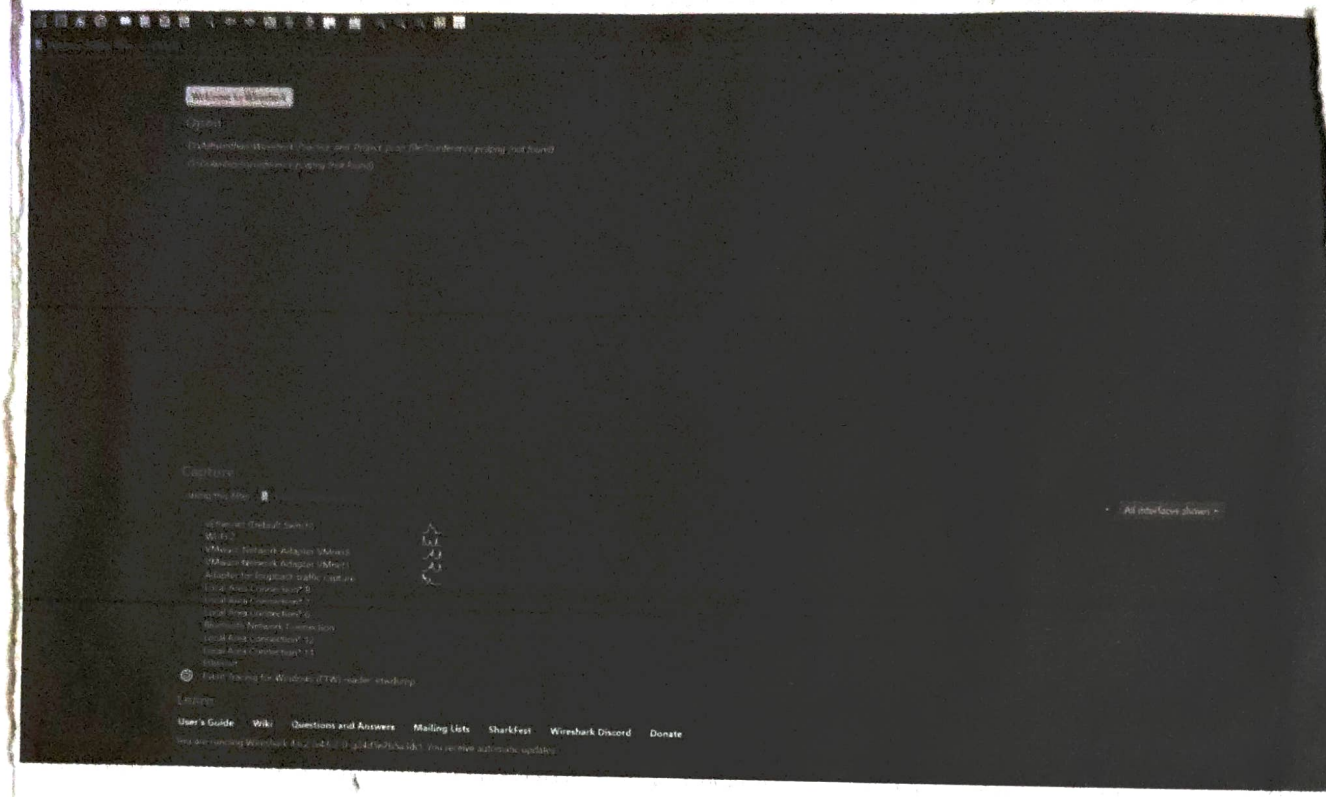
i) Create a filter to display only TCP/UDP packets, inspect the packets and provide the flow graph

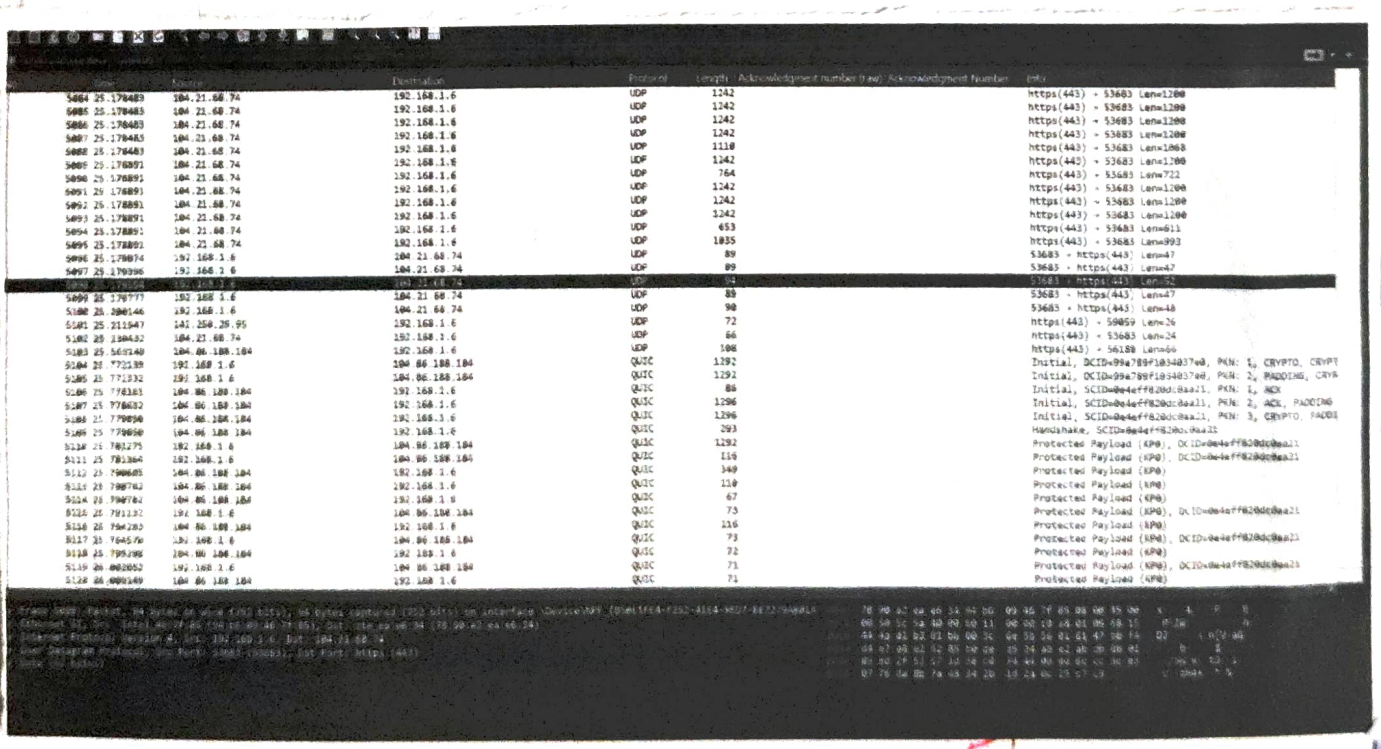
→ Go to search bar at the top of the tool

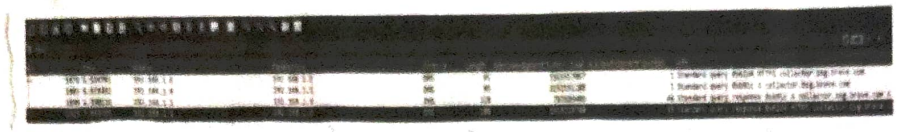
→ Search TCP packets in search bar.

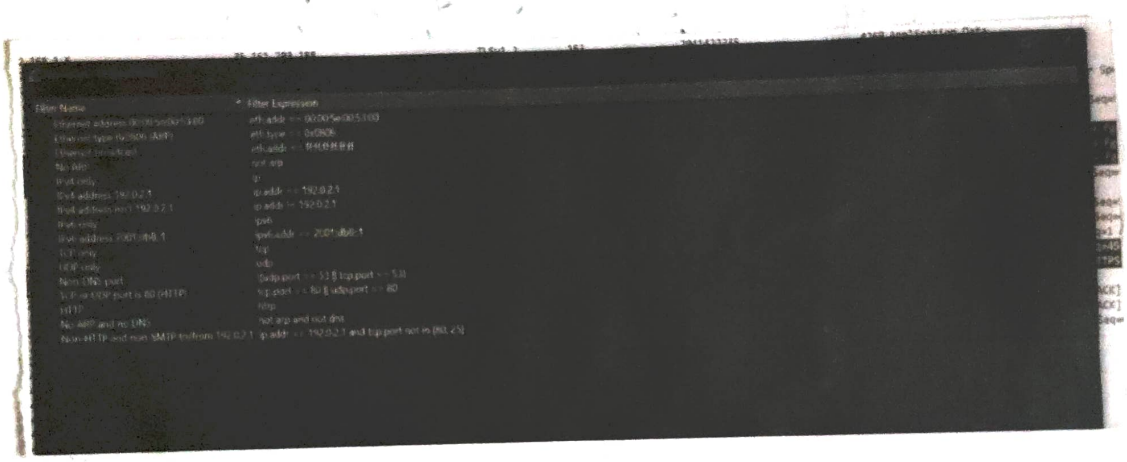
→ To see flow graph click statistics flow graph.

→ Save the packets


Welcome to Wireshark
Capture
Learn
User's Guide
Wiki
Questions and Answers
Mailing Lists
SharkFest
Wireshark Discord
Donate


No. Time Source Destination Protocol Length Acknowledgment number (raw) Acknowledgment number Info
5084 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5085 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5086 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5087 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5088 25.178483 104.21.68.74 192.168.1.6 UDP 1110 https(443) → 53683 Len=1068
5089 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5090 25.178891 104.21.68.74 192.168.1.6 UDP 764 https(443) → 53683 Len=722
5091 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5092 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5093 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5094 25.178891 104.21.68.74 192.168.1.6 UDP 653 https(443) → 53683 Len=611
5095 25.178891 104.21.68.74 192.168.1.6 UDP 1035 https(443) → 53683 Len=993
5096 25.178974 192.168.1.6 104.21.68.74 UDP 89 53683 → https(443) Len=47
5097 25.179396 192.168.1.6 104.21.68.74 UDP 89 53683 → https(443) Len=47
5098 25.179558 192.168.1.6 104.21.68.74 UDP 94 53683 → https(443) Len=52
5099 25.179777 192.168.1.6 104.21.68.74 UDP 89 53683 → https(443) Len=47
5100 25.200146 192.168.1.6 104.21.68.74 UDP 90 53683 → https(443) Len=48
5101 25.211547 142.250.29.95 192.168.1.6 UDP 72 https(443) → 59059 Len=26
5102 25.230432 104.21.68.74 192.168.1.6 UDP 66 https(443) → 53683 Len=24
5103 25.569140 104.86.188.184 192.168.1.6 UDP 106 https(443) → 56188 Len=66
5104 25.772139 192.168.1.6 104.86.188.184 QUIC 1292 Initial, DCID=99a789f1034837e0, PKN: 1, CRYPTO, CRYPT
5105 25.772332 192.168.1.6 104.86.188.184 QUIC 1292 Initial, DCID=99a789f1034837e0, PKN: 2, PADDING, CRYP
5106 25.776383 104.86.188.184 192.168.1.6 QUIC 86 Initial, SCID=0e4eff820dc8aa21, PKN: 1, ACK
5107 25.776632 104.86.188.184 192.168.1.6 QUIC 1296 Initial, SCID=0e4eff820dc8aa21, PKN: 2, ACK, PADDING
5108 25.779050 104.86.188.184 192.168.1.6 QUIC 1296 Initial, SCID=0e4eff820dc8aa21, PKN: 3, CRYPTO, PADDI
5109 25.779050 104.86.188.184 192.168.1.6 QUIC 293 Handshake, SCID=0e4eff820dc8aa21
5110 25.781275 192.168.1.6 104.86.188.184 QUIC 1292 Protected Payload (KP0), DCID=0e4eff820dc8aa21
5111 25.781364 192.168.1.6 104.86.188.184 QUIC 116 Protected Payload (KP0), DCID=0e4eff820dc8aa21
5112 25.790603 104.86.188.184 192.168.1.6 QUIC 349 Protected Payload (KP0)
5113 25.790782 104.86.188.184 192.168.1.6 QUIC 110 Protected Payload (KP0)
5114 25.790782 104.86.188.184 192.168.1.6 QUIC 67 Protected Payload (KP0)
5115 25.791232 192.168.1.6 104.86.188.184 QUIC 73 Protected Payload (KP0), DCID=0e4eff820dc8aa21
5116 25.794283 104.86.188.184 192.168.1.6 QUIC 116 Protected Payload (KP0)
5117 25.794570 192.168.1.6 104.86.188.184 QUIC 73 Protected Payload (KP0), DCID=0e4eff820dc8aa21
5118 25.795198 104.86.188.184 192.168.1.6 QUIC 72 Protected Payload (KP0)
5119 26.002052 192.168.1.6 104.86.188.184 QUIC 71 Protected Payload (KP0), DCID=0e4eff820dc8aa21
5120 26.000149 104.86.188.184 192.168.1.6 QUIC 71 Protected Payload (KP0)




Filter Name
Filter Expression
arp
ip
ip.addr == 192.0.2.1
ip.addr != 192.0.2.1
ipv6
ipv6.addr == 2001:db8::1
tcp
udp
tcp.port == 80 || udp.port == 80
http
not arp and not dns
ip.addr == 192.0.2.1 and tcp.port not in {80 443}



i) Create a filter to display DNS packets and provide

ii) Create a filter for ARP packets

* Go to search option
* Search ARP packet in search bar.
* Save the packets.

iii) Creating a filter to display DNS packets and provide the flow graph.

iv) Creating a filter for ICMP and DHCP packets and inspect and show the packets

v) Create filter for ICMP and DHCP packets and inspect and show the packets.

RESULT:-

Wireshark helps us in analysing and understanding our network traffic and packet flow through various filters and flow graphs.